Deep Reinforcement Learning-Based Downlink Beamforming and Phase Optimization for RIS-Aided Communication System

Lingjie Li, *Member, IEEE*, Yang Yang, *Member, IEEE*, Lingyan Bao, Zhen Gao, *Member, IEEE*, Yongpeng Wu, *Senior Member, IEEE*, and Honglin Xiang

***Abstract*—Reconfigurable intelligent surface (RIS) has been envisioned as a critical technology for the future wireless communication systems. However, the inter-cell interference and the unbearable signal processing complexity seriously constrain the wide application of RIS. In this letter, we investigate a deep reinforcement learning (DRL)-based scheme of downlink beamforming and phase optimization for a multi-cell RIS-aided communication system. With the transmit power constraint, a maximization problem of average ergodic rate is formulated. We design a deep deterministic policy gradient (DDPG)-based scheme, which enables base station (BS) and RIS to intelligently control the beamforming and phase shift, respectively. Further, a DDPG-based distributed training decentralized execution algorithm is proposed to acquire the feature difference among multi-cells and maximize the average ergodic rate. Simulation results show that the proposed algorithm outperforms the other baseline algorithms, which proves the agents can effectively learn and sense the change of multi-cell interference.**

***Index Terms*—Beamforming, deep reinforcement learning, inter-cell interference, phase shift, reconfigurable intelligent surface.**

I. Introduction

RECONFIGURABLE intelligent surface (RIS) is envisioned as a critical technology for future wireless communication systems [1]. By controlling programmable electromagnetic (EM) units on a planar surface structure, RIS is expected to break through the uncontrollability of traditional wireless channels and build an intelligent programmable wireless environment [2].

Some previous studies have been committed to improving the performance of the RIS-aided single-cell communication systems [3], [4]. Reference [5] studied joint beamforming coordination and resource allocation in the downlink multi-cell systems. Further, an iterative convex approximation (ICA) algorithm [6] and a distributed successive concave approximation (D-SCA) algorithm [7] were proposed to improve the performance of the RIS-aided multi-cell systems, respectively. Nevertheless, these traditional methods require perfect channel state information (CSI) and incur non-negligible execution time due to the estimation of high-dimensional cascaded CSI and calculation of beamformers and phase shifts, which may not be effective to the practical environment where the channels are dynamic.

Fortunately, deep reinforcement learning (DRL) provides a unique paradigm to solve these problems effectively. Huang et al. [8] first developed a DRL-based algorithm to optimize the phase shift at the RIS. On this basis, both deep learning and DRL are used in [9] to jointly optimize the power allocation of one single base station (BS) and the phase shift of the RIS. DRL is beneficial to dynamic wireless communication systems, since it allows systems to learn and build knowledge about the channels without knowing the channel model and mobility pattern. However, the above-mentioned studies assumed perfect global CSI and did not consider the joint optimization of beamforming and phase shift in a dynamic multi-cell system with partial observability of the CSI.

In this letter, we propose a DRL-based method of downlink beamforming and phase shift optimization for the RIS-aided multi-cell communication system. Based on DRL, we define an agent to jointly control the beam of each BS and the phase shift of RIS, respectively. With the transmit power constraint, a maximization problem of average ergodic rate is formulated and this problem is non-convex due to the uncertainty of the other cell interference. Thus, we design a deep deterministic policy gradient (DDPG)-based distributed training decentralized execution (DTDE) algorithm. In our scheme, each agent trains its own DDPG by the exchange of information and executes the action in a distributed manner, which avoids the mutual misinformation between agents. Simulation results show that compared with other benchmarks, our proposed DDPG-based DTDE algorithm achieves the best performance and converges much faster than the deep Q network (DQN)-based algorithm, which proves the agents can effectively learn and sense the change of multi-cell interference in the dynamic RIS-aided communication system.

The rest of this letter is organized as follows. Section II introduces the system model. Section III demonstrates the DDPG-based downlink beamforming and phase optimization, and Section IV presents the DDPG-based DTDE algorithm. Finally, simulation results are illustrated in Section V, and Section VI concludes this letter.

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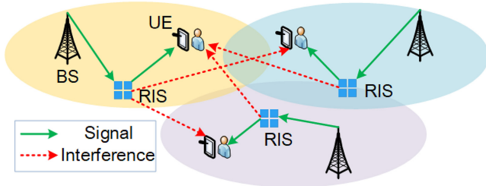

Fig. 1. System model.

II. System Model

We consider a downlink RIS-aided communication system with K cells. Each cell consists of an M-antenna BS, multiple single-antenna user equipments (UEs) and a RIS equipped with N reflecting elements as shown in Fig. 1. We assume that the direct signal transmissions are insignificant. It is reasonable since RIS is generally deployed to overcome the situations where signal transmissions are blocked by obstacles. In each cell, there exists a cascaded BS-RIS-UE channel. The signals from BS first arrive at RIS, and then by intelligently programming to vary phase shifts, the signals are reflected to the UEs. The UEs served by a BS in a cell use orthogonal frequency bands, so there is no intra-cell interference. However, there exists inter-cell interference. To facilitate the analysis, we focus on one downlink with RIS in the k-th cell $(k = 1, 2, \dots, K)$ by assigning only one user in each cell, and the other cell j $(j \neq k,\ j = 1, 2, \dots, K)$ shares the same frequency resources with cell k. The flat-and-block fading channel model is adopted. Specifically, the downlink channels between RIS k and UE k and between the BS k and RIS k at time t are modeled, respectively. Which are shown as follows:

$$\mathbf{h}_{R_k,U_k}(t) = \sqrt{\gamma_k}\mathbf{g}_{R_k,U_k}\ \mathbf{A}(N, \psi_{DoD}, \Delta) \in \mathbb{C}^{1\times N}, \quad (1)$$

$$\mathbf{H}_{B_k,R_k}(t) = \sqrt{\beta_k}\mathbf{G}_{B_k,R_k}\ \mathbf{A}(M, \theta_{DoD}, \Delta) \in \mathbb{C}^{N\times M}, \quad (2)$$

where γ_k and β_k denote the large-scale fading coefficient consisting of the path loss and shadowing from the RIS to the UE and from the BS to the RIS, respectively; $\mathbf{g}_{R_k,U_k} \in \mathbb{C}^{1\times L}$ is the Rayleigh fading vector composed of the small scale fading coefficients of all the multi-paths and $\mathbf{G}_{B_k,R_k} \in \mathbb{C}^{N\times L}$ consists of N Rayleigh fading vector from the BS to the n-th reflecting element; L denotes the total number of multi-paths. The Rayleigh fading vector is a complex Gaussian distribution, and it remains constant in each time slot and varies between successive time slots according to the first-order Gaussian–Markov process [10]. Thus, the channels are time-varying. $\mathbf{A}(N, \psi_{DoD}, \Delta) \in \mathbb{C}^{L\times N}$ and $\mathbf{A}(M, \theta_{DoD}, \Delta) \in \mathbb{C}^{L\times M}$ are the array response matrices composed of the steering vectors of all multi-paths, where ψ_{DoD} and θ_{DoD} denotes the nominal direction of departure (DoD) of the channel between RIS and UE and between BS and RIS, respectively; Δ denotes the angular spread.

Based on the practical RIS model, the phase shift applied at the reflecting RIS k at time t can be written as $\mathbf{\Phi}_k(t) \triangleq \mathrm{diag}\,[\phi_1, \phi_2, \dots, \phi_N] \in \mathbb{C}^{N\times N}$, $\mathbf{\Phi}_k(t)$ is a diagonal matrix whose principal diagonal elements are given by $\phi_n = e^{j\varphi_n}$, where $\varphi_n \in [0, 2\pi)$ is the phase shift introduced by the n-th reflecting element. Moreover, we assume $|\phi_n|^2 = 1$ to ensure an ideal reflection without signal power loss.

The received signal of UE k at time slot t is written as:

$$y_k(t) = \mathbf{h}_{R_k,U_k}(t)\mathbf{\Phi}_k(t)\mathbf{H}_{B_k,R_k}(t)\mathbf{w}_k(t)x_k(t) + \sum_{j\neq k}^{K}\mathbf{h}_{R_j,U_k}(t)\mathbf{\Phi}_j(t)\mathbf{H}_{B_j,R_j}(t)\mathbf{w}_j(t)x_j(t) + z_k(t), \quad (3)$$

where $w_k(t) \in \mathbb{C}^{M\times 1}$ denotes the beamformer of UE k; $x_k(t)$ denotes the signal transmitted to the UE k; $z_k(t) \sim \mathcal{N}(0, \sigma^2)$ denotes the additive white complex Gaussian noise with zero-mean and variance σ^2. Therefore, the signal-to-interference-plus-noise ratio (SINR) of UE k at time slot t is given by

$$\Upsilon_k(\mathbf{W}(t), \mathbf{\Phi}(t)) = \frac{|\mathbf{h}_{R_k,U_k}(t)\mathbf{\Phi}_k(t)\mathbf{H}_{B_k,R_k}(t)\mathbf{w}_k(t)|^2}{|\sum_{j\neq k}^{K}\mathbf{h}_{R_j,U_k}(t)\mathbf{\Phi}_j(t)\mathbf{H}_{B_j,R_j}(t)\mathbf{w}_j(t)|^2 + \sigma^2}, \quad (4)$$

where $\mathbf{W}(t) = [\mathbf{w}_1(t), \mathbf{w}_2(t), \dots, \mathbf{w}_K(t)]$, $\mathbf{\Phi}(t) = [\mathbf{\Phi}_1(t), \mathbf{\Phi}_2(t), \dots, \mathbf{\Phi}_K(t)]$.

Then, we aim to maximize the average ergodic rate of the multi-cell RIS-aided communication system by optimizing the beamforming matrix $\mathbf{W}(t)$ and the configuration matrix $\mathbf{\Phi}(t)$ under the transmit power constraint P_t. Thus, the resulting optimization problem can be formulated as

$$\begin{aligned} &\max_{\mathbf{W}(t),\mathbf{\Phi}(t)} \frac{\sum_{k=1}^{K} R_k(\mathbf{W}(t), \mathbf{\Phi}(t))}{K}, \\ &\text{s.t.} \quad tr\left\{\mathbf{W}\mathbf{W}^T\right\} = P_t, \\ &|\phi_n|^2 = 1, \forall n = 1, 2, \dots N, \end{aligned} \quad (5)$$

where $R_k(\mathbf{W}(t), \mathbf{\Phi}(t))$ is the ergodic rate of the k-th cell, which can be written as:

$$R_k(\mathbf{W}(t), \mathbf{\Phi}(t)) = \log(1 + \Upsilon_k(\mathbf{W}(t), \mathbf{\Phi}(t))). \quad (6)$$

III. DDPG-Based Downlink Beamforming and Phase Optimization for RIS-Aided System

A standard architecture of DRL consists of four key elements: agent, action, reward and policy. The optimization problem can be viewed as a Markov Decision process (MDP) problem. In each step, the agent consistently observes the state s_t of the environment. With the observed s_t, the agent takes action a_t by policy $\pi(a \mid s)$ and interacts with the environment, and then will receive the reward r_t and the new state s_{t+1}.

The design of the state, action and reward are described in details as follows.

State: Local information $\mathcal{S}_l$: The local information is comprised of seven elements. The first two elements are the action taken by the BS k in time slot $t-1$, namely, *1)* the beamformer vector $\mathbf{c}_k(t-1)$ and *2)* the phase shift matrix $\mathbf{\Phi}_k(t-1)$. Since the neural network input values cannot be a matrix, we input the phase shift φ_n introduced by the n-th reflecting element in the neural network. The third element is *3)* the UE's ergodic rate of cell k in time slot $t-1$, i.e., $R_k(\mathbf{W}(t-1), \mathbf{\Phi}(t-1))$). To capture the effect of channel variation over time, the fourth and fifth elements are the channel gains in two consecutive time slot t and $t-1$: *4)* $|\mathbf{h}_{R_k,U_k}(t-1)\mathbf{\Phi}_k(t-2)\mathbf{H}_{B_k,R_k}(t-1)\mathbf{c}_k(t-2)|^2$ and *5)* $|\mathbf{h}_{R_k,U_k}(t)\ \mathbf{\Phi}_k(t-1)\ \mathbf{H}_{B_k,R_k}(t)\mathbf{c}_k(t-1)|^2$. The sixth and seventh elements are the total interference-plus-noise power received by BS k: *6)* $|\sum_{j\neq k}^{K}\mathbf{h}_{R_j,U_k}(t)\mathbf{\Phi}_j(t-1)\mathbf{H}_{B_j,R_j}(t)\mathbf{w}_j(t-1)|^2 + \sigma^2$, *7)* $|\sum_{j\neq k}^{K}\mathbf{h}_{R_j,U_k}(t-1)\mathbf{\Phi}_j(t-2)\mathbf{H}_{B_j,R,R_j}(t-1)\mathbf{w}_j(t-2)|^2 + \sigma^2$. The number of entries formed here is $6+N$.

Neighborhood information: The signal transmitted from BS k to UE k will interfere with and be interfered by neighboring cells. Therefore, we divide the neighborhood information into interferers' information $\mathcal{U}_k(t)$ and interfered neighbors' information $\mathcal{V}_k(t)$. We only consider the top i interference in order from largest to smallest, i.e., $|\mathcal{U}_k(t)| = i$ and $|\mathcal{V}_k(t)| = i$.

For interferer x, there exists five elements in interferers' information $\mathcal{S}_{\mathcal{U}}$. The first two elements are *1)* the index (i.e., x), *2)* the beamformer vector index of BS x which make the interference. The third element is *3)* the phase shift of RIS x. The fourth element is *4)* the interference by the BS x: $|\mathbf{h}_{R_j,U_k}(t)\mathbf{\Phi}_j(t)\mathbf{H}_{B_j,R_j}(t)\mathbf{w}_j(t)|^2$. The last element is *5)* the ergodic rate of the UE x: $R_x(\mathbf{W}(t), \mathbf{\Phi}(t))$. We consider i interferers' information in the current and past time slots, thus there will be $(8 + 2N) \cdot i$ entries of states. For interfered neighbor y, there exist three elements in interfered neighbors' information $\mathcal{S}_{\mathcal{V}}$: *1)* $R_y(\mathbf{W}(t-1), \mathbf{\Phi}(t-1))$, *2)* $|\mathbf{h}_{R_y,U_y}(t-1)\mathbf{\Phi}_y(t-1)\mathbf{H}_{B_y,R_y}(t-1)\mathbf{c}_y(t-1)|^2$ and *3)* $\frac{|\mathbf{h}_{R_k,U_y}(t)\mathbf{\Phi}_k(t)\mathbf{H}_{B_k,R_k}(t)\mathbf{w}_k(t)|^2}{|\sum_{j\neq y}\mathbf{h}_{R_i,U_y}(t)\mathbf{\Phi}_i(t)\mathbf{H}_{B_i,R_i}(t)\mathbf{w}_i(t)|^2+\sigma^2}$. The number of entries of the state constructed from interfered neighbors is $3i$.

Therefore, the states set can be represented as follows:

$$\mathcal{S} = \mathcal{S}_l \cup \mathcal{S}_{\mathcal{U}} \cup \mathcal{S}_{\mathcal{V}}. \tag{7}$$

The dimension of the state space is $D_s = (2N + 11) \cdot i + N + 6$.

Action: The action space is continuous and comprised of the transmit beamformer $\mathbf{w}_k(t) = \sqrt{p} \cdot \mathbf{c}_k(t)$ and the phase shift matrix $\mathbf{\Phi}_k(t)$ of cell k at time slot t, where p denotes the power of BS and $\mathbf{c}_k(t)$ denotes the direction of the beam. The actor network can directly generate the n-th phase shift ϕ_n by $\phi_n = e^{j\varphi_n}$. We consider a codebook $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_Q] \in \mathbb{C}^{M\times Q}$ composed of Q optional beams, which is designed as follows. $\mathbf{C}[m, q]$ denotes the phase shift of the m-th $(m = 1, 2, \ldots, M)$ antenna element in the q-th code.

$$\mathbf{C}[m, q] = \frac{1}{\sqrt{M}} \exp\left(j\frac{2\pi}{S}\left\lfloor \frac{m * \mathrm{mod}(q + \frac{Q}{2}, Q)}{Q/S} \right\rfloor\right), \tag{8}$$

where the S denotes the number of available phase values for each antenna element; $\lfloor\cdot\rfloor$ denotes the floor function; Q denotes the number of beam patterns.

Therefore, the actions set can be represented as follows:

$$\mathcal{A} = \{(\varphi_1, \varphi_2, \ldots, \varphi_N, c_i), c_i \in [1, Q]\}, \tag{9}$$

where the action variable c_i needs to be discretized to the index of the beam c'_i, i.e., if $c_i = 0$, then $c'_i = 1$; if $c_i \neq 0$, then $c'_i = \lceil c_i \rceil$, where $\lceil\cdot\rceil$ is the ceiling operation.

The dimension of the action space is $D_a = N + 1$.

Reward: When one agent chooses an action to maximize its ergodic rate, it will generate high interference to neighbored cells and cause a severe decline of the system's performance. Therefore, we design a reward function, which not only maximizes the ergodic rate of cell k but also minimizes the negative influence of its action on the neighbored cells. The penalty on agent k is defined as the sum of the ergodic rate losses of the neighbors $y \in \mathcal{V}_k(t+1)$ interfered by BS k,

$$\mathrm{P}_k(\mathbf{W}(t), \mathbf{\Phi}(t)) = \sum_{y\in\mathcal{V}_k(t+1)} -C_y(\mathbf{W}(t), \mathbf{\Phi}(t))$$

$$+\log\left(1 + \frac{|\mathbf{h}_{R_y,U_y}(t)\mathbf{\Phi}_y(t)\mathbf{H}_{B_y,R_y}(t)\mathbf{w}_y(t)|^2}{|\sum_{i\neq k,y}\mathbf{h}_{R_i,U_y}(t)\mathbf{\Phi}_i(t)\mathbf{H}_{B_i,R_i}(t)\mathbf{w}_i(t)|^2 + \sigma^2}\right). \tag{10}$$

Setting the scalar factor of the penalty term to one avoids the agent behaving selfishly and choosing suboptimal actions for fear of punishment. The reward function is designed as:

$$r_k(t) = R_k(\mathbf{W}(t), \mathbf{\Phi}(t)) - \mathrm{P}_k(\mathbf{W}(t), \mathbf{\Phi}(t)). \tag{11}$$

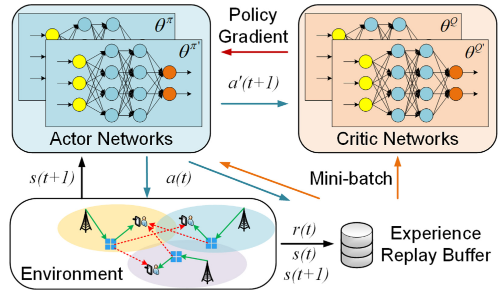

Fig. 2. Diagram of the proposed DDPG-based DTDE algorithm.

IV. DDPG-Based DTDE Algorithm

Our letter proposes a multi-agent DDPG framework of DTDE. Assuming the BS in each cell is able to constantly collect the information of the environment, and each BS is regarded as an independent agent which trains its own DDPG. An agent's state and reward depend not only on its own action but also on the actions taken by others. Therefore, the Markov property is violated, and the convergence guarantees of single agent are no longer valid. One solution is a centralized training decentralized execution (CTDE) scheme. In a CTDE scheme, there exists a centralized controller which can collect information about all agents. After centralized learning, the decentralized policies are executed according to local observations, which leads to mutual misinformation between agents. In our proposed DTDE scheme, this problem is addressed with information exchange among agents and the design of the reward function. Historical actions, channel measurements, and ergodic rates are shared between the agents instead of the parameters of networks to learn the impact of their coupled actions on the dynamic environment. In addition, we design a reward function to avoid the agents behaving selfishly and help them make appropriate decisions.

Most of the prior works are based on statistical CSI or perfect instantaneous CSI. Statistical CSI requires both channel mean and channel variance matrix, which are obtained from the instantaneous CSI [11]. Also, the performance with it is generally worse [12]. Moreover, the cascaded channel and the channel dynamics severely limit the channel estimation accuracy. In cellular networks, the BSs transmit pilot symbols and then the UEs measure the downlink scalar channel gains and feedback to the BSs [13]. In our proposed algorithm, BSs only need to acquire and exchange scalar channel gains instead of the high-dimensional complex-valued matrices in conventional algorithms, which significantly reduces the signaling overhead.

Our proposed DDPG-based DTDE algorithm is shown in Fig. 2. In the online execution stage, the agent k (BS k) executes action a_t according to its current state s_t, which is obtained from UEs and the information exchanged with other BSs, and its trained policy. In the offline training stage, we first initialize

Algorithm 1 DDPG-Based DTDE Algorithm

Initialize the parameters of networks: θ^Q, θ^π, $\theta^{Q'}$, $\theta^{\pi'}$.
Initialize the replay buffer with size M, sample size N_{batch}.
Repeat
 Agent k observes its current state s_t;
 Agent k chooses action $\pi_k(a \mid s) + \lambda\rho$ and executes;
 Agent k obtains the new state $s(t+1)$ and reward r_t;
 Agent k stores the experience (s_t, a_t, r_t, s_{t+1}) into the replay buffer;
 if the learning process starts **then**
  Agent k randomly chooses N_{batch} experiences from the replay buffer;
  Update the critic network by the loss function;
  Update the actor network by the policy gradient;
  Soft update the target networks;
 end if
Until convergence

the parameters of networks and the replace buffer separately. According to the current state s_t and the policy $\pi_k(a \mid s)$, agent k chooses action a_t added a random noise ρ which decays with rate λ for exploration and then gets the reward r_t and the new state s_{t+1}. Then, the experience (s_t, a_t, r_t, s_{t+1}) is stored into the replay buffer. When the learning process starts, agent k randomly chooses mini-batch experiences to train. The training process is shown in **Algorithm 1**. The critic network is updated by minimizing the loss function:

$$\frac{\partial L(\theta^Q)}{\partial \theta^Q} = E_{s,a,r,s'\sim N}\left[\left(r + \gamma Q'\left(s', \pi\left(s'\middle|\theta^{\pi'}\right)\middle|\theta^{Q'}\right) - Q(s, a \mid \theta^Q)\right)\frac{\partial Q\left(s, a\middle|\theta^Q\right)}{\partial \theta^Q}\right]. \tag{12}$$

The actor network is trained with policy gradient:

$$\nabla_{\theta^\pi} J \approx \frac{1}{N}\sum_i (\nabla_a Q(s_i, a \mid \theta^\pi)|_{a=\pi(s_i)} \cdot \nabla_{\theta^\pi}\pi(s_i \mid \theta^\pi)). \tag{13}$$

The two target networks are update by rate τ:

$$\theta^{Q'} = \tau\theta^Q + (1-\tau)\theta^{Q'}, \tag{14}$$
$$\theta^{\pi'} = \tau\theta^\pi + (1-\tau)\theta^{\pi'}. \tag{15}$$

Typically, for a dense neural network, the complexity is $\mathcal{O}(\nu\mu^2)$, where ν is the number of layers and μ is the number of neurons for the widest layer, i.e., the layer with the most neurons [5]. In our case, μ depends on the dimension of the input layer, i.e., $\mathcal{O}(\mu) = \mathcal{O}((2N+11)\cdot i + N + 6)$. The number of layers for a neural network is independent of the scale of the problem. Since the total number of agents is K, the computational complexity of our proposed algorithm is $\mathcal{O}(K\cdot((2N+11)\cdot i + N + 6)^2)$.

V. Simulation Evaluation

In this section, simulations are provided to evaluate the performance of the proposed algorithm for the multi-cell RIS-aided communication system. We consider that there is a cell in the center and two layers of cells outside, with 6 and 12 cells, respectively. In each cell, BS is located in the center and RIS is deployed between the BS and UE. The cell radius is 200m and the transmit power of BS is set as 38dBm. The large-scale fading coefficient can be modeled as $\gamma_k = \beta_k = 120.9 + 37.6\log_{10} d$, where d denotes the distance of the path. The number of multi-paths L is 4, the angular spread Δ is 3° and the noise power is –114dBm. The number of reflecting elements of RIS is set to 4. For the design of the codebook matrix, we set $S = 16$, $M = 3$ and $Q = 4$.

TABLE I
Hyper-Parameters Descriptions

Parameter	Description	Value
T	total number of training steps	80000
M	buffer size of experience replay	500
N_{batch}	the number of mini-batch	32
γ	discount factor	0.5
θ^Q,$\theta^{Q'}$	learning rate of critic network	0.0001
θ^π,$\theta^{\pi'}$	learning rate of actor network	0.0001
ε	ε-greedy policy factor	$[0.01, 0.6]$
τ	soft update rate	0.01

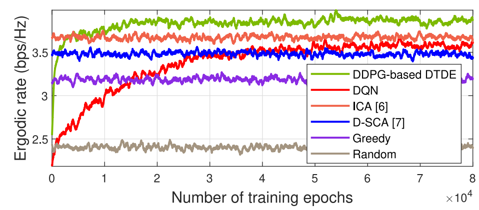

Fig. 3. Average ergodic rate of six different algorithms.

In our proposed algorithm, all the neural networks are DNNs using three fully-connected hidden layers with 400, 300 and 10 neurons, respectively. All the activation functions of the DNNs are *ReLU*, except for the output layer of actor network, which uses the *tanh* instead of *sigmoid* to mitigate the vanishing gradient problem. The elements in the actions set are obtained by adding one to the output values of the neural network and then scaling. We employ the Adam optimizer for training. The hyperparameters for simulation are shown in Table I.

We adopt five benchmarks for comparison. The first one is DQN-based algorithm, which uses the same codebook as the DDPG-based algorithm, i.e., $S = 16$, $M = 3$ and $Q = 4$. We assume the phase shift introduced by the n-th reflecting element can be selected from a discrete domain, i.e., $\varphi_n \in \{0, \Delta\varphi, \ldots, (K-1)\Delta\varphi\}$, where $\Delta\varphi = 2\pi/K$. We set K as 4 and each agent trains its own DQN network. The second is ICA-based algorithm of [6] and the third is D-SCA-based algorithm of [7]. The fourth is Greedy-based algorithm, in which each agent obtains the action by following a greedy approach. The last is Random-based algorithm, where the beam pattern and phase shift of RIS are randomly selected.

As illustrated in Fig. 3, in a dynamic wireless environment, the proposed DDPG-based algorithm achieves best performance. It not only converges much faster but also achieves a higher average ergodic rate than the DQN-based algorithm since the latter can only provide the designed phase shift in several levels. The execution times of DDPG, DQN, ICA and D-SCA based algorithm are approximately 0.71ms, 0.82ms, 12000ms and 11000ms, respectively.

In Section IV, we consider the top i interference information. With the increase of i, agents can exchange more information and get more valuable observations, which leads to better decisions. As shown in Fig. 4, we investigate the impact of i and the different codebook size on the performance of our proposed algorithm. As i increases, the average ergodic

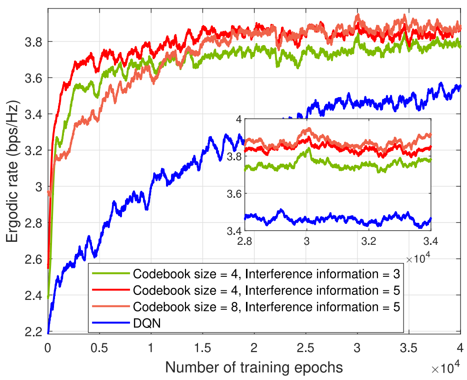

Fig. 4. Average ergodic rate for different number of codebook size and interference information.

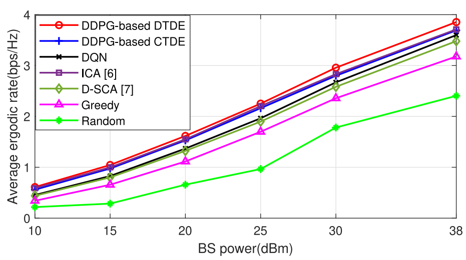

Fig. 5. Average ergodic rate vs. BS power.

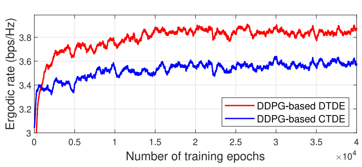

Fig. 6. DTDE vs. CTDE.

TABLE II
CHOSEN BEAMFORMING PATTERNS

	Pattern1	Pattern2	Pattern3	Pattern4
DDPG-based CTDE	17	0	0	2
DDPG-based DTDE	8	2	4	5

rate also increases as expected. Also, it can also be observed that the algorithm with codebook size equals 8 achieves better performance than algorithm with codebook size equals 4.

We investigate the impact of BS power on our proposed algorithms as illustrated in Fig. 5. Since the results of all algorithms have small fluctuations due to the dynamic wireless environment in Fig. 3, we take the average of 10,000 epochs after the convergence of DRL-based algorithms as the average ergodic rate. We can see that the average ergodic rate increases with the power. As the BS power increases, the SINR of each cell also increases. Thus, the system can achieve a higher average ergodic rate. Moreover, with joint design of beamforming and phase shift, the proposed algorithm achieves the best performance for different BS power levels.

It can be seen in Fig. 6 that the DTDE algorithm converges slower than the CTDE algorithm due to the lack of mutual information. Also, training a public network from the experiences of all agents in a CTDE algorithm means that an agent could be misled by others. As shown in Table II, we take two snapshots of chosen beamforming patterns of CTDE and DTDE algorithms in a same time slot after they get saturated. In a CTDE algorithm, most of the BSs (17 of 19) tend to choose the same pattern because the agents make a decision by the DDPGs with the same weights. However, the agent performs much better performance at choosing the pattern which suits best in the DTDE algorithm. The execution time of CTDE algorithm and DTDE algorithm is approximately 0.71ms and 0.70ms, respectively. The agents in DTDE algorithm output the actions in parallel and the number of layers of the network is small and the structure is not complex, so the difference in execution time is very small.

VI. CONCLUSION

In this letter, we consider a downlink multi-cell RIS-aided communication system. Specifically, a DDPG-based scheme is proposed to extract features of the neighborhood interference information, which enables BS and RIS to intelligently obtain an optimal design strategy of beamforming and phase shift. Further, through the interaction between agents, a DDPG-based DTDE algorithm is proposed to maximize the average ergodic rate of the RIS-aided system. Simulation results show that the proposed algorithm not only converges much faster than the DQN-based algorithm, but also achieves better performance than the other five benchmarks.

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